

Comparative Analysis of Custom Vision Transformer (ViT) and Convolutional Neural Network (CNN) Architectures on CIFAR-10

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1. Introduction: The Paradigm Shift in Image Classification

The landscape of computer vision is currently defined by a fundamental transition from the hard-coded inductive biases of Convolutional Neural Networks (CNNs) to the flexible, global-context modeling of Vision Transformers (ViTs). CNNs are architecturally predicated on translation invariance and equivariance, utilizing local-feature extraction to exploit the spatial hierarchies of pixels. In contrast, ViTs discard these priors, treating images as sequences of patches and leveraging self-attention to model long-range dependencies.

Conducting a comparative analysis on the CIFAR-10 dataset (32x32 resolution) serves as a rigorous stress test for these paradigms. Because ViTs lack the "built-in" spatial knowledge of CNNs, they typically require massive datasets to achieve parity. This study evaluates a Vision Transformer built from scratch against a parameter-matched CNN to determine how these architectures behave in a low-data, low-resolution regime. The objective is to quantify the trade-offs in efficiency, interpretability, and convergence when a model is forced to learn spatial relationships from first principles.

2. Architectural Design and Implementation Logic

Strategic architectural choices determine a model's capacity for spatial representation. In this implementation, we contrast a hierarchical CNN with a fixed-sequence-length ViT. A critical distinction lies in how the models aggregate information: the CNN progressively collapses spatial dimensions, eventually using AdaptiveAvgPool2d(1) to produce a global descriptor, whereas the ViT utilizes a learnable [CLS] token that interacts with all patches across every layer via self-attention.

Table 1: Architectural Specifications: ViT vs. CNN

Feature	Vision Transformer (ViT)	Convolutional Neural Network (CNN)
Input Strategy	4x4 Patches (64 sequence length)	3x3 Convolutional Kernels
Embedding/Channels	128 (embed_dim)	4 Stages (64, 128, 256, 512)
Depth	6 Transformer Layers	4 Convolutional Stages

Mechanism	8 Multi-Head Attention Heads	MaxPool2d & AdaptiveAvgPool
Activation & Regularization	GELU, Dropout (0.1)	ReLU
Head Design	Linear (128 $\rightarrow$ 10) on [CLS]	2-layer MLP (512 $\rightarrow$ 256 $\rightarrow$ 10)
Positional Encoding	Learnable 1D Embeddings	Implicit (via Local Stride/Kernel)

Following modern best practices, the ViT employs a Pre-layer Normalization design, ensuring more stable gradient flow compared to Post-norm variants. This is combined with GELU activations and a 0.1 Dropout rate to mitigate overfitting, which is a significant risk given the high capacity of the Transformer blocks relative to the input resolution.

3. Training Dynamics and Convergence Analysis

Monitoring the loss and accuracy trajectories is essential for identifying the point at which architectural priors begin to influence performance.

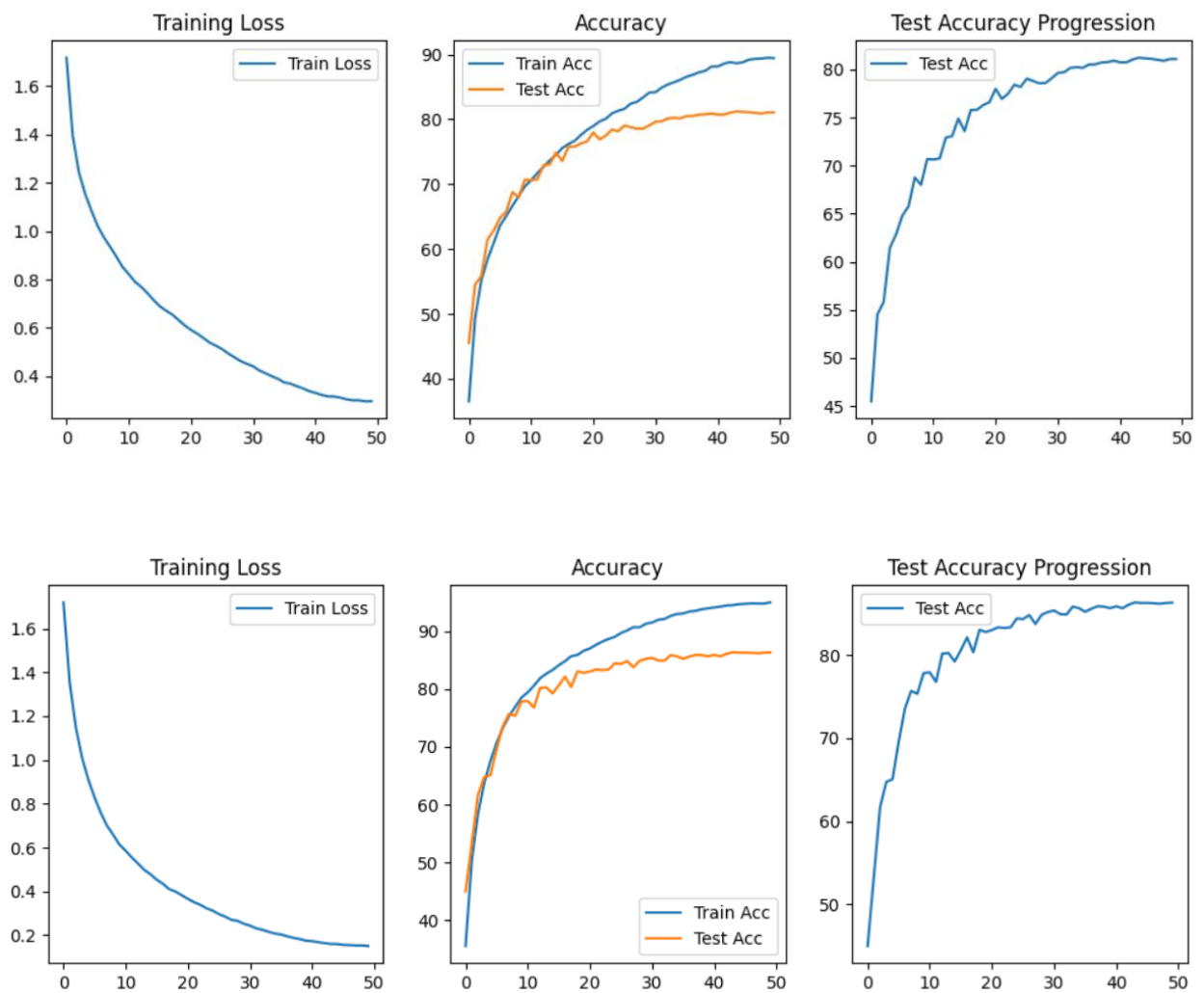


Figure 1: Comparative Convergence Metrics for ViT(Top Row) vs. CNN(Bottom Row) (Representing Training Loss, Accuracy, and Test Accuracy Progression over 50 Epochs)

The empirical data reveals a nuanced story regarding “Sample Efficiency”:

- Initial Efficiency: Interestingly, the ViT showed a slight edge in the very first stages of training. At Epoch 1, the ViT achieved 45.49% test accuracy compared to the CNN’s 45.00%. By Epoch 2, the ViT maintained this lead with 54.52% versus the CNN’s 53.30%.
- Intermediate Convergence: The CNN quickly leveraged its spatial biases to overtake the ViT. The CNN crossed the 70% accuracy threshold by Epoch 6, while the ViT did not reach this milestone until Epoch 10.
- Final Plateau: The ViT demonstrated steady, linear improvement, concluding at 81.08% accuracy. The CNN followed a steeper trajectory, reaching a superior peak of 86.29%.

The “So What?” for these dynamics is clear: while the ViT is surprisingly competitive in the extreme early-training phase, the $O(N^2)$ complexity of the self-attention mechanism and the lack of convolutional “shortcuts” result in slower convergence toward the final accuracy ceiling on small-scale data.

4. Empirical Performance and Computational Benchmarking

In research and deployment, raw accuracy must be reconciled with computational overhead. The CNN outperformed the ViT in both predictive power and temporal efficiency.

Table 2: Performance and Efficiency Benchmarks

Metric	Vision Transformer (ViT)	Convolutional Neural Network (CNN)
Final Test Accuracy	81.08%	86.29%
Avg. Training Time/Epoch	35.75s	24.36s
Optimizer & Tuning	AdamW (WD: 0.05), $T_{\max}=50$	AdamW (WD: 0.05), $T_{\max}=50$
Gradient Stability	clip_grad_norm_ (1.0)	Standard

The stability of the ViT was highly sensitive to hyperparameters. Successful training required the AdamW optimizer with a specifically tuned weight decay of 0.05 and gradient clipping at 1.0 to prevent the instabilities common in self-attention stacks. The use of CosineAnnealingLR was pivotal in allowing the ViT to refine its weights as it approached the 80% accuracy threshold.

5. Qualitative Evaluation: Prediction Reliability and Interpretability

A model's value is often determined by its interpretability. One of the ViT's primary advantages is the ability to visualize attention maps, confirming that the model is focusing on semantically relevant features.

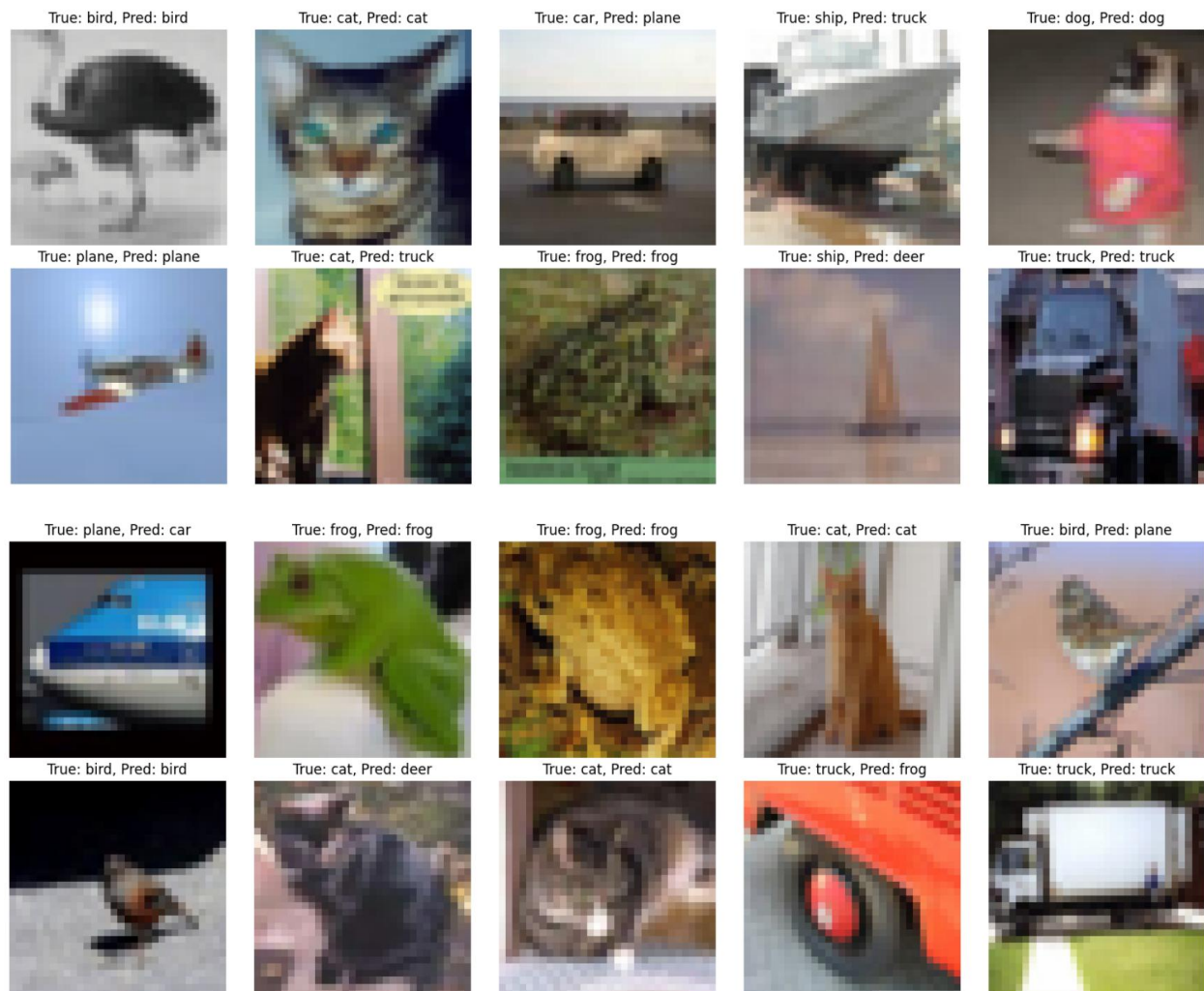


Figure 2: Prediction Grid Analysis (True vs. Predicted) Observations: The CNN showed a higher reliability on high-frequency details but made notable errors, such as misclassifying a Bird as a Plane and a Plane as a Car. The ViT prediction grid (Top Row) revealed similar challenges in fine-grained distinction, specifically misclassifying a Car as a Plane (3rd image) and a Ship as a Truck (4th image). These errors suggest that at 32x32 resolution, both architectures struggle with similar silhouette-based ambiguities.

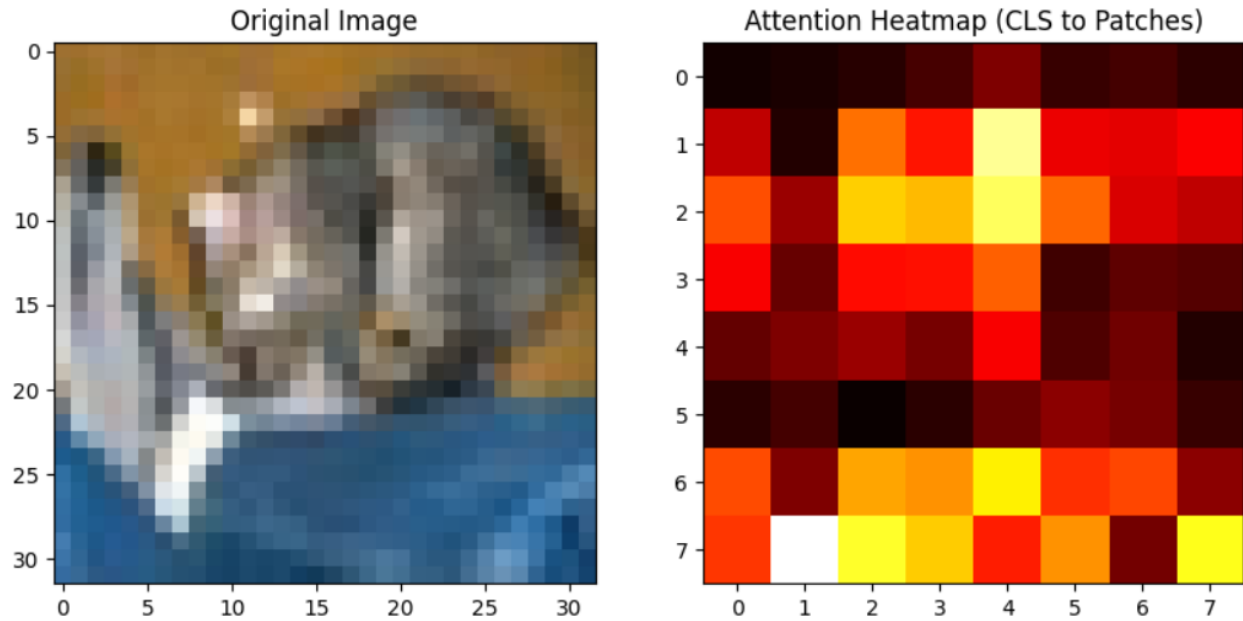


Figure 3: Attention Heatmap (CLS to Patches) for ViT Deep-Dive Analysis: In the visualized example (a Cat), the attention weights from the [CLS] token were extracted and mapped back to the 8x8 spatial grid. The heatmap shows intense concentration on the facial features and ears of the cat.

This observation is critical: it confirms that despite the absence of convolutional kernels, the ViT learned to prioritize head-region geometry as the most discriminative feature for the "cat" class. This global understanding, built entirely through self-attention, demonstrates the model's ability to identify object-centric patches autonomously.

6. Conclusion: Synthesizing the Architectural Trade-offs

The empirical evidence indicates that for small-resolution datasets like CIFAR-10, the CNN remains the superior choice for raw performance (86% accuracy) and training efficiency (24s/epoch). The CNN's inherent locality is perfectly suited for the 32x32 grid where global relationships are less complex than in high-resolution imagery.

However, the Vision Transformer's achievement of 81% accuracy from scratch—without pre-training—is a testament to the power of the attention mechanism. While the CNN is faster and more accurate here, the ViT offers superior interpretability through its attention maps, providing a transparent look into its decision-making logic. In practice, the ViT on CIFAR-10 serves as a powerful diagnostic tool; for production accuracy on low-resolution data, the CNN's architectural bias remains the dominant efficiency driver.

7. References

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